AERIS Expert Review Packet

Anomaly

Source / metric	tceq / no2
Observed / expected	7.5 / 1.1223
Time	2026-07-08 06:00 UTC
Location	29.5831, -95.0155
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	humidity	percent	6 / 620	39.12 to 100; mean 72.707	88.7 at 2026-07-08 05:55Z; dt 5 min
asos	temperature	degC	6 / 620	23 to 36.1111; mean 29.005	25 at 2026-07-08 05:55Z; dt 5 min
asos	wind_direction	degree	6 / 714	0 to 340; mean 132.143	0 at 2026-07-08 05:55Z; dt 5 min
asos	wind_speed	m/s	6 / 748	0 to 10.8033; mean 2.3288	0 at 2026-07-08 05:55Z; dt 5 min
noaa_gfs	gh_500	m	6 / 78	5896.72 to 5924.75; mean 5911.54	5918.79 at 2026-07-08 06:00Z; dt 0 min
noaa_gfs	pbl_height	m	6 / 78	28.4394 to 2153.73; mean 762.601	327.872 at 2026-07-08 06:00Z; dt 0 min
noaa_gfs	precipitable_water	mm	6 / 78	39.1305 to 51.7657; mean 46.5426	46.5186 at 2026-07-08 06:00Z; dt 0 min
noaa_gfs	surface_pressure	hPa	6 / 78	1011.03 to 1017.68; mean 1014.99	1017.3 at 2026-07-08 06:00Z; dt 0 min
noaa_gfs	t_850	degC	6 / 78	17.233 to 20.4926; mean 18.9262	19.6123 at 2026-07-08 06:00Z; dt 0 min
noaa_gfs	u_10m	m/s	6 / 78	-2.38144 to 1.93714; mean 0.123	0.61561 at 2026-07-08 06:00Z; dt 0 min
noaa_gfs	v_10m	m/s	6 / 78	-0.854414 to 4.60966; mean 2.1594	2.71584 at 2026-07-08 06:00Z; dt 0 min
openaq	ozone	ppm	15 / 672	-0.004 to 0.069; mean 0.0175	0.005 at 2026-07-08 06:00Z; dt 0 min
openaq	pm10	ug/m3	3 / 142	8 to 66; mean 25.0423	16 at 2026-07-08 06:00Z; dt 0 min
openaq	pm25	ug/m3	67 / 2372	0 to 50.7; mean 5.2015	4.6 at 2026-07-08 06:00Z; dt 0 min
openweather	cloud_cover	percent	3 / 216	0 to 100; mean 23.4028	13 at 2026-07-08 06:01Z; dt 1.9 min
openweather	humidity	percent	3 / 216	38 to 95; mean 76.2361	86 at 2026-07-08 06:01Z; dt 1.9 min
openweather	precipitation	mm/h	3 / 216	0 to 5.62; mean 0.0692	0 at 2026-07-08 06:01Z; dt 1.9 min
openweather	pressure	hPa	3 / 216	1012 to 1019; mean 1016.44	1018 at 2026-07-08 06:01Z; dt 1.9 min
openweather	temperature	degC	3 / 216	24.19 to 35.68; mean 29.6379	26.06 at 2026-07-08 06:01Z; dt 1.9 min
openweather	wind_direction	degree	3 / 216	0 to 359; mean 197.569	205 at 2026-07-08 06:01Z; dt 1.9 min
openweather	wind_speed	m/s	3 / 216	0.36 to 8; mean 2.3444	2.57 at 2026-07-08 06:01Z; dt 1.9 min
purpleair	pm25	ug/m3	60 / 4143	0 to 146.4; mean 7.2069	0.4 at 2026-07-08 05:59Z; dt 0 min
sentinel5p	s5p_aer_ai_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-07-07 20:37Z; dt 562.4 min

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
sentinel5p	s5p_ch4_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-07-07 20:37Z; dt 562.4 min
sentinel5p	s5p_co_column	mol/m^2	6 / 6	0.0243723 to 0.0253077; mean 0.0249	0.0249873 at 2026-07-07 18:57Z; dt 662.4 min
sentinel5p	s5p_co_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-07-07 20:37Z; dt 562.4 min
sentinel5p	s5p_hcho_column	mol/m^2	8 / 8	0.000174097 to 0.000235341; mean 0.0002	0.000202795 at 2026-07-07 20:37Z; dt 562.4 min
sentinel5p	s5p_hcho_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-07-07 20:37Z; dt 562.4 min
sentinel5p	s5p_no2_column	mol/m^2	4 / 4	3.63106e-05 to 4.30727e-05; mean 0	4.10314e-05 at 2026-07-07 20:37Z; dt 562.4 min
sentinel5p	s5p_no2_granule_available	count	4 / 4	1 to 1; mean 1	1 at 2026-07-07 20:37Z; dt 562.4 min
sentinel5p	s5p_o3_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-07-07 20:37Z; dt 562.4 min
sentinel5p	s5p_so2_column	mol/m^2	8 / 8	-3.49304e-05 to 4.72775e-05; mean 0	2.51087e-05 at 2026-07-07 20:37Z; dt 562.4 min
sentinel5p	s5p_so2_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-07-07 20:37Z; dt 562.4 min
tceq	co	ppm	3 / 163	0 to 1; mean 0.235	0.2 at 2026-07-08 06:00Z; dt 0 min
tceq	no2	ppb	11 / 578	-0.1 to 35.8; mean 6.2433	7.5 at 2026-07-08 06:00Z; dt 0 min
tceq	so2	ppb	3 / 154	-0.5 to 6.8; mean -0.0747	-0.2 at 2026-07-08 06:00Z; dt 0 min

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	humidity	07-06 18Z 55.13, 07-07 00Z 76.52, 06Z 88.01, 12Z 69.83, 18Z 53.58, 07-08 00Z 78.5, 06Z 90.93, 12Z 69.98, 18Z 58.27, 07-09 00Z 76.88, 06Z 88.36, 12Z 67.8
asos	temperature	07-06 18Z 32.77, 07-07 00Z 28.48, 06Z 26.02, 12Z 29.96, 18Z 32.88, 07-08 00Z 26.09, 06Z 24.77, 12Z 30.39, 18Z 31.92, 07-09 00Z 28.28, 06Z 25.73, 12Z 30.43
asos	wind_direction	07-06 18Z 168.9, 07-07 00Z 129.8, 06Z 70.98, 12Z 128.9, 18Z 164.8, 07-08 00Z 116.1, 06Z 23.62, 12Z 182.7, 18Z 169.8, 07-09 00Z 146.4, 06Z 118.3, 12Z 160.7
asos	wind_speed	07-06 18Z 4.084, 07-07 00Z 2.151, 06Z 0.844, 12Z 1.479, 18Z 3.523, 07-08 00Z 2.864, 06Z 0.3568, 12Z 2.121, 18Z 4.013, 07-09 00Z 2.643, 06Z 1.444, 12Z 2.387
noaa_gfs	gh_500	07-06 18Z 5903, 07-07 00Z 5899, 06Z 5905, 12Z 5899, 18Z 5914, 07-08 00Z 5912, 06Z 5918, 12Z 5912, 18Z 5924, 07-09 00Z 5914, 06Z 5919, 12Z 5910, 18Z 5921
noaa_gfs	pbl_height	07-06 18Z 1742, 07-07 00Z 788.1, 06Z 171.2, 12Z 77.48, 18Z 1802, 07-08 00Z 824.4, 06Z 201.3, 12Z 100.2, 18Z 1948, 07-09 00Z 214.1, 06Z 247.5, 12Z 113.4, 18Z 1684
noaa_gfs	precipitable_water	07-06 18Z 43.91, 07-07 00Z 47.6, 06Z 45.54, 12Z 46.61, 18Z 49.19, 07-08 00Z 49.19, 06Z 46.64, 12Z 46.81, 18Z 48.62, 07-09 00Z 45.51, 06Z 44.36, 12Z 44.64, 18Z 46.43
noaa_gfs	surface_pressure	07-06 18Z 1014, 07-07 00Z 1012, 06Z 1014, 12Z 1015, 18Z 1016, 07-08 00Z 1014, 06Z 1016, 12Z 1017, 18Z 1016, 07-09 00Z 1014, 06Z 1015, 12Z 1015, 18Z 1015
noaa_gfs	t_850	07-06 18Z 18.1, 07-07 00Z 19.9, 06Z 19.74, 12Z 18.83, 18Z 17.67, 07-08 00Z 19.17, 06Z 19.63, 12Z 19.03, 18Z 18.03, 07-09 00Z 19.89, 06Z 19.42, 12Z 18.96, 18Z 17.67
noaa_gfs	u_10m	07-06 18Z 0.2464, 07-07 00Z -1.198, 06Z 0.3297, 12Z 1.134, 18Z -0.8133, 07-08 00Z -1.616, 06Z 0.4156, 12Z 1.132, 18Z 0.4076, 07-09 00Z -0.07952, 06Z 1.055, 12Z 1.149, 18Z -0.5631
noaa_gfs	v_10m	07-06 18Z 1.711, 07-07 00Z 4.026, 06Z 2.07, 12Z 0.8139, 18Z 1.497, 07-08 00Z 4.036, 06Z 2.383, 12Z 0.8573, 18Z 0.8856, 07-09 00Z 3.191, 06Z 2.597, 12Z 1.426, 18Z 2.579

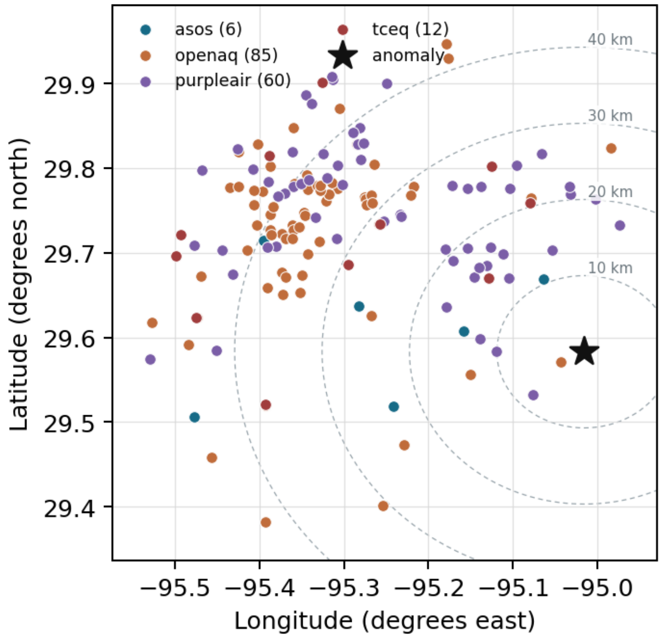
Source	Metric	Values
openaq	ozone	07-06 18Z 0.03379, 07-07 00Z 0.01325, 06Z 0.004926, 12Z 0.01279, 18Z 0.02943, 07-08 00Z 0.02414, 06Z 0.004, 12Z 0.004111, 18Z 0.035, 07-09 12Z 0.01635, 18Z 0.02708
openaq	pm10	07-06 18Z 27.5, 07-07 00Z 15.89, 06Z 18.28, 12Z 33.17, 18Z 29.88, 07-08 00Z 21.94, 06Z 24.39, 12Z 36, 18Z 23, 07-09 12Z 27.25, 18Z 25
openaq	pm25	07-06 18Z 5.265, 07-07 00Z 4.241, 06Z 5.95, 12Z 6.585, 18Z 7.067, 07-08 00Z 4.338, 06Z 6.606, 12Z 5.775, 18Z 3.028, 07-09 00Z 3.561, 06Z 4.493, 12Z 4.918, 18Z 1.919
openweather	cloud_cover	07-06 18Z 24.61, 07-07 00Z 5.842, 06Z 2.444, 12Z 19.44, 18Z 50.22, 07-08 00Z 36.28, 06Z 4.611, 12Z 3.765, 18Z 40.47, 07-09 00Z 70.16, 06Z 18.83, 12Z 2.389
openweather	humidity	07-06 18Z 61.67, 07-07 00Z 76.58, 06Z 87.67, 12Z 76.33, 18Z 60.94, 07-08 00Z 77.89, 06Z 91.28, 12Z 78.35, 18Z 62.71, 07-09 00Z 75.95, 06Z 88.67, 12Z 76.17
openweather	precipitation	07-06 18Z 0.008889, 07-07 00Z 0, 06Z 0, 12Z 0, 18Z 0, 07-08 00Z 0.7911, 06Z 0, 12Z 0, 18Z 0.03235, 07-09 00Z 0, 06Z 0, 12Z 0
openweather	pressure	07-06 18Z 1014, 07-07 00Z 1014, 06Z 1016, 12Z 1018, 18Z 1016, 07-08 00Z 1016, 06Z 1017, 12Z 1019, 18Z 1017, 07-09 00Z 1016, 06Z 1016, 12Z 1018
openweather	temperature	07-06 18Z 34.19, 07-07 00Z 29.43, 06Z 26.47, 12Z 30.28, 18Z 34.05, 07-08 00Z 27.8, 06Z 24.8, 12Z 30.05, 18Z 33.25, 07-09 00Z 29.06, 06Z 26.26, 12Z 30.29
openweather	wind_direction	07-06 18Z 180.4, 07-07 00Z 190.4, 06Z 212.7, 12Z 206.8, 18Z 186.3, 07-08 00Z 169.6, 06Z 201.8, 12Z 239.4, 18Z 192.8, 07-09 00Z 177.8, 06Z 205.6, 12Z 210.8
openweather	wind_speed	07-06 18Z 2.329, 07-07 00Z 2.734, 06Z 1.768, 12Z 1.452, 18Z 2.861, 07-08 00Z 2.998, 06Z 2.081, 12Z 2.04, 18Z 2.951, 07-09 00Z 3.045, 06Z 2.073, 12Z 1.756
purpleair	pm25	07-06 18Z 6.49, 07-07 00Z 6.121, 06Z 7.782, 12Z 8.078, 18Z 8.724, 07-08 00Z 7.736, 06Z 9.811, 12Z 7.884, 18Z 5.593, 07-09 00Z 5.09, 06Z 6.826, 12Z 6.278, 18Z 4.5
sentinel5p	s5p_aer_ai_granule_available	07-06 18Z 1, 07-07 18Z 1, 07-08 18Z 1
sentinel5p	s5p_ch4_granule_available	07-06 18Z 1, 07-07 18Z 1, 07-08 18Z 1
sentinel5p	s5p_co_column	07-06 18Z 0.02528, 07-07 18Z 0.02495, 07-08 18Z 0.02437
sentinel5p	s5p_co_granule_available	07-06 18Z 1, 07-07 18Z 1, 07-08 18Z 1
sentinel5p	s5p_hcho_column	07-06 18Z 0.0002254, 07-07 18Z 0.0002152, 07-08 18Z 0.000185
sentinel5p	s5p_hcho_granule_available	07-06 18Z 1, 07-07 18Z 1, 07-08 18Z 1
sentinel5p	s5p_no2_column	07-06 18Z 4.086e-05, 07-07 18Z 3.867e-05, 07-08 18Z 4.307e-05
sentinel5p	s5p_no2_granule_available	07-06 18Z 1, 07-07 18Z 1, 07-08 18Z 1
sentinel5p	s5p_o3_granule_available	07-06 18Z 1, 07-07 18Z 1, 07-08 18Z 1
sentinel5p	s5p_so2_column	07-06 18Z 4.559e-05, 07-07 18Z 2.628e-06, 07-08 18Z -2.084e-05
sentinel5p	s5p_so2_granule_available	07-06 18Z 1, 07-07 18Z 1, 07-08 18Z 1
tceq	co	07-06 18Z 0.1882, 07-07 00Z 0.23, 06Z 0.25, 12Z 0.2167, 18Z 0.2706, 07-08 00Z 0.2, 06Z 0.2444, 12Z 0.25, 07-09 06Z 0.2222, 12Z 0.2722, 18Z 0.1667
tceq	no2	07-06 18Z 4.021, 07-07 00Z 3.418, 06Z 5.972, 12Z 5.721, 18Z 7.052, 07-08 00Z 6.258, 06Z 10.67, 12Z 7.524, 18Z 1.3, 07-09 06Z 5.853, 12Z 5.429, 18Z 4.273
tceq	so2	07-06 18Z 0.075, 07-07 00Z -0.23, 06Z -0.2687, 12Z 0.3333, 18Z -0.05294, 07-08 00Z -0.2, 06Z -0.1867, 12Z -0.03889, 07-09 06Z -0.1556, 12Z -0.09412, 18Z -0.1667

Per-site averages

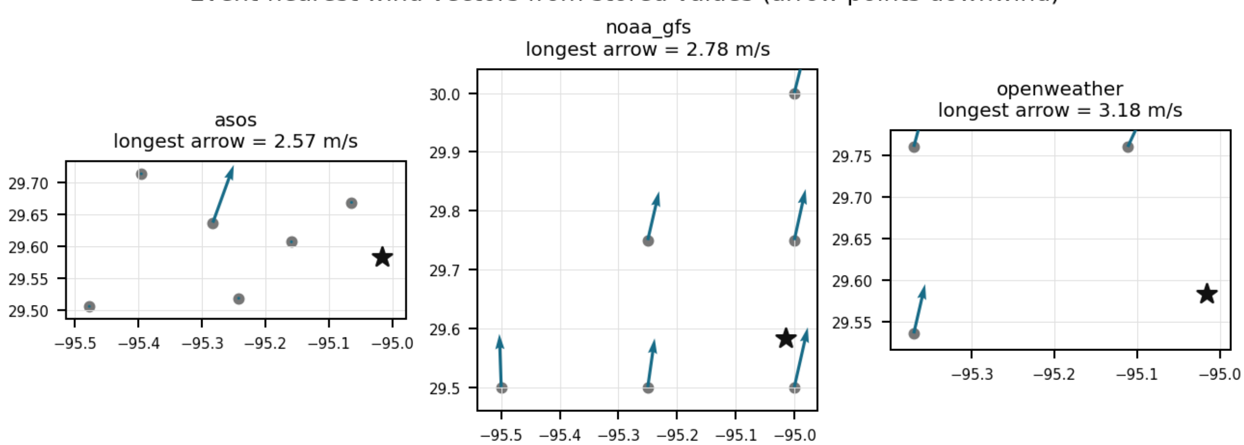
Each metric averaged at each site, with that site's distance from the event in brackets.

Source	Metric	Values
asos	humidity	T41 (10.667 km) 75.15, EFD (14.102 km) 68.48, LVJ (23.01 km) 71.63, HOU (26.499 km) 72.75, MCJ (39.455 km) 72.3, AXH (45.443 km) 72.84
asos	temperature	T41 (10.667 km) 28.88, EFD (14.102 km) 30.38, LVJ (23.01 km) 29.21, HOU (26.499 km) 29.3, MCJ (39.455 km) 29.01, AXH (45.443 km) 28.33
asos	wind_direction	T41 (10.667 km) 161.1, EFD (14.102 km) 145.3, LVJ (23.01 km) 116.4, HOU (26.499 km) 161.6, MCJ (39.455 km) 139.7, AXH (45.443 km) 78.92
asos	wind_speed	T41 (10.667 km) 2.714, EFD (14.102 km) 3.281, LVJ (23.01 km) 1.859, HOU (26.499 km) 2.87, MCJ (39.455 km) 2.897, AXH (45.443 km) 0.9475
noaa_gfs	gh_500	gfs:29.50,-95.00 (9.357 km) 5912, gfs:29.75,-95.00 (18.624 km) 5911, gfs:29.50,-95.25 (24.49 km) 5912, gfs:29.75,-95.25 (29.288 km) 5911, gfs:30.00,-95.00 (46.387 km) 5911, gfs:29.50,-95.50 (47.768 km) 5912
noaa_gfs	pbl_height	gfs:29.50,-95.00 (9.357 km) 709, gfs:29.75,-95.00 (18.624 km) 832, gfs:29.50,-95.25 (24.49 km) 711.8, gfs:29.75,-95.25 (29.288 km) 949.8, gfs:30.00,-95.00 (46.387 km) 621.1, gfs:29.50,-95.50 (47.768 km) 752
noaa_gfs	precipitable_water	gfs:29.50,-95.00 (9.357 km) 47.06, gfs:29.75,-95.00 (18.624 km) 47.17, gfs:29.50,-95.25 (24.49 km) 46.26, gfs:29.75,-95.25 (29.288 km) 46.59, gfs:30.00,-95.00 (46.387 km) 46.85, gfs:29.50,-95.50 (47.768 km) 45.33
noaa_gfs	surface_pressure	gfs:29.50,-95.00 (9.357 km) 1016, gfs:29.75,-95.00 (18.624 km) 1016, gfs:29.50,-95.25 (24.49 km) 1015, gfs:29.75,-95.25 (29.288 km) 1015, gfs:30.00,-95.00 (46.387 km) 1014, gfs:29.50,-95.50 (47.768 km) 1014
noaa_gfs	t_850	gfs:29.50,-95.00 (9.357 km) 19, gfs:29.75,-95.00 (18.624 km) 18.99, gfs:29.50,-95.25 (24.49 km) 18.93, gfs:29.75,-95.25 (29.288 km) 18.94, gfs:30.00,-95.00 (46.387 km) 18.79, gfs:29.50,-95.50 (47.768 km) 18.91
noaa_gfs	u_10m	gfs:29.50,-95.00 (9.357 km) 0.01572, gfs:29.75,-95.00 (18.624 km) -0.05505, gfs:29.50,-95.25 (24.49 km) 0.1519, gfs:29.75,-95.25 (29.288 km) 0.3149, gfs:30.00,-95.00 (46.387 km) 0.2365, gfs:29.50,-95.50 (47.768 km) 0.07418
noaa_gfs	v_10m	gfs:29.50,-95.00 (9.357 km) 2.27, gfs:29.75,-95.00 (18.624 km) 2.116, gfs:29.50,-95.25 (24.49 km) 2.035, gfs:29.75,-95.25 (29.288 km) 2.298, gfs:30.00,-95.00 (46.387 km) 2.101, gfs:29.50,-95.50 (47.768 km) 2.136
openaq	ozone	5077791 (0.008 km) 0.01475, 332 (14.586 km) 0.02089, 2800 (21.051 km) 0.0194, 320 (24.786 km) 0.01511, 3214 (26.617 km) 0.01986, 339 (26.892 km) 0.02294, 2039 (28.563 km) 0.01724, 3378 (28.774 km) 0.01522, +7 more
openaq	pm10	271 (14.586 km) 16.92, 13380082 (28.774 km) 30.4, 15852419 (35.786 km) 28.63
openaq	pm25	14404954 (2.969 km) 3.213, 13955815 (13.307 km) 5.581, 268 (14.586 km) 6.864, 11638043 (23.95 km) 2.294, 14157975 (24.789 km) 7.409, 3379 (28.774 km) 7.95, 15539682 (28.776 km) 6.419, 8967123 (28.83 km) 7.132, +59 more
openweather	cloud_cover	grid:east (21.774 km) 23.18, grid:south (34.665 km) 22.78, grid:center (39.501 km) 24.25
openweather	humidity	grid:east (21.774 km) 78.76, grid:south (34.665 km) 74.32, grid:center (39.501 km) 75.62
openweather	precipitation	grid:east (21.774 km) 0.1236, grid:south (34.665 km) 0.001944, grid:center (39.501 km) 0.08208
openweather	pressure	grid:east (21.774 km) 1017, grid:south (34.665 km) 1016, grid:center (39.501 km) 1016
openweather	temperature	grid:east (21.774 km) 29.56, grid:south (34.665 km) 29.51, grid:center (39.501 km) 29.84
openweather	wind_direction	grid:east (21.774 km) 192.3, grid:south (34.665 km) 199.3, grid:center (39.501 km) 201.2
openweather	wind_speed	grid:east (21.774 km) 2.877, grid:south (34.665 km) 2.115, grid:center (39.501 km) 2.041
purpleair	pm25	2386 (8.161 km) 0.1083, 3298 (10.11 km) 6.319, 247283 (12.095 km) 5.894, 268165 (12.925 km) 7.207, 222815 (13.892 km) 5.254, 222593 (15.865 km) 6.61, 128007 (15.888 km) 1.293, 161689 (15.943 km) 5.31, +52 more
tceq	co	482011039 (14.566 km) 0.1836, 482011035 (28.771 km) 0.1173, 482011052 (44.211 km) 0.3946
tceq	no2	482011050 (0.0 km) 1.507, 482011039 (14.566 km) 4.472, 482011015 (20.507 km) 7.156, 482010026 (26.633 km) 7.887, 482011035 (28.771 km) 10.71, 482010416 (29.325 km) 4.802, 480391004 (37.123 km) 2.664, 482011052 (44.211 km) 14.32, +3 more
tceq	so2	482011039 (14.566 km) 0.03125, 482011035 (28.771 km) -0.07692, 482010051 (44.59 km) -0.1667

Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly



Event-nearest wind vectors from stored values (arrow points downwind)



Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). The labeling guide that comes with this packet has the definitions and marking rules. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Claim 1 of 35

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5 particles.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 2 of 35

Surface and model data support unusually poor dispersion at the anomaly time because the nearest ASOS observation reported 0.0 m/s wind with 88.7 percent humidity and about 25.0 degC near 05:55 UTC, while GFS indicated a shallow planetary boundary layer near 328 m and surface pressure near 1017 hPa at 06:00 UTC.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 3 of 35

The temperature inversion over the Houston area is causing a buildup of pollutants near the surface.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 4 of 35

The wind patterns in the area are not effectively dispersing pollutants, leading to a concentration of air pollutants near the anomaly location.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 5 of 35

Nighttime stagnation and suppressed mixing were a likely primary physical cause of the NO_2 anomaly because the anomaly occurred with near-calm surface wind at the nearest ASOS station, very low 6-hour mean wind speeds across ASOS sites, and a shallow nighttime planetary boundary layer in GFS.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 6 of 35

A very shallow nocturnal planetary boundary layer combined with near-zero surface wind speeds trapped ground-level pollutants near the surface around 06:00 UTC.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 7 of 35

ASOS surface weather observations near the anomaly site recorded calm wind speeds of 0.0 m/s at 05:55 UTC on July 8, 2026.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 8 of 35

Fresh local combustion emissions from traffic, diesel equipment, marine activity, or other fuel-burning sources near the monitor were a likely cause of the NO2 anomaly because NO2 rose strongly while ozone was very low at the same time, a pattern characteristic of recently emitted NOx in nighttime conditions.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 9 of 35

The air quality anomaly in Houston, Texas is caused by high levels of SO2 particles.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 10 of 35

A severe NO2 concentration anomaly of 7.5 ppb was detected by TCEQ monitor 482011050 at coordinates (29.583, -95.016) on 2026-07-08T06:00:00+00:00, representing a z-score of 6.88 above the expected value of 1.12 ppb.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 11 of 35

Regional chemistry data are more consistent with fresh NOx accumulation than with a broad secondary pollution episode because openaq ozone was very low near 0.005 ppm at 06:00 UTC while TCEQ area NO2 6-hour mean rose to 10.67 ppb during 07-08 06Z.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 12 of 35

The PM2.5 levels are elevated at 0.4 ug/m3, which is higher than the mean of 7.2069 ug/m3.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 13 of 35

Regional TCEQ monitors observed a peak 6-hour average NO_2 concentration of 10.67 ppb during the period centered at 2026-07-08T06:00:00+00:00.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 14 of 35

The air quality anomaly in Houston, Texas is also characterized by elevated levels of NO_2.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 15 of 35

A narrow source plume from the Houston Ship Channel or nearby industrial facilities is a plausible contributing cause because the anomalous monitor sits in an industrialized corridor and the local wind field around the anomaly time was weak and variable enough to allow a short-lived plume to impact one site more strongly than the regional background.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 16 of 35

The tceq NO2 monitor at 29.583, -95.016 recorded 7.5 ppb at 2026-07-08T06:00:00+00:00, compared with an expected value of 1.1222972972972973 ppb, yielding a z-score of 6.8822187328850974 and a severe anomaly classification.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 17 of 35

The 6-hour mean of tceq NO2 across 11 monitors was 10.67 ppb for 2026-07-08 06Z, which was higher than the preceding 2026-07-08 00Z mean of 6.258 ppb.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 18 of 35

The tceq NO2 anomaly was identified by three detection methods: zscore, stl, and isolation_forest.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 19 of 35

TCEQ monitoring showing baseline levels for carbon monoxide and sulfur dioxide during the 06:00 UTC window on July 8, 2026, contradicts the hypothesis of an uncharacteristically large industrial explosion or major regional fire event.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 20 of 35

A nearby industrial facility or power plant is releasing excessive amounts of particulate matter and gases into the atmosphere.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 21 of 35

The absence of daytime photolysis coupled with chemical titration of ambient ozone supported the nighttime accumulation of NO2.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 22 of 35

OpenAQ ambient ozone concentrations falling to a 6-hour mean of 0.004 ppm around 06:00 UTC on July 8, 2026, supports nocturnal ozone titration by fresh nitrogen oxide emissions as an auxiliary chemical mechanism raising TCEQ NO2 levels.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 23 of 35

The air quality anomaly in Houston, Texas is characterized by elevated levels of PM2.5.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 24 of 35

NOAA GFS boundary layer heights dropping to near 200 meters alongside ASOS surface wind speeds decreasing to near 0.35 m/s at 06:00 UTC on July 8, 2026, provide strong supporting evidence for atmospheric stagnation and nocturnal surface layer trapping of localized NO2 emissions detected by TCEQ.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 25 of 35

The air quality anomaly in Houston, Texas is likely caused by a combination of factors including industrial activities and vehicle emissions.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 26 of 35

Continuous local anthropogenic emissions from industrial facilities and transport corridors accumulated near the monitoring site under calm atmospheric conditions.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 27 of 35

The air quality anomaly is related to PM2.5 levels.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 28 of 35

Fresh nearby combustion emissions likely contributed to the NO2 spike because the TCEQ NO2 6-hour mean rose to 10.67 ppb during 07-08 06Z while ozone from nearby OpenAQ sensors was very low at 0.005 ppm at 06:00 UTC and about 0.004 ppm for the 6-hour mean, a pattern consistent with nighttime fresh NOx influence and ozone titration, although TCEQ CO was only modest at 0.2 ppm at 06:00 UTC and 0.2444 ppm for the 6-hour mean.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 29 of 35

The most likely primary cause of the 2026-07-08 06:00 UTC NO2 anomaly at the TCEQ monitor near 29.583, -95.016 is nighttime trapping of local combustion emissions under stagnant conditions.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 30 of 35

NOAA GFS model data indicated a shallow nocturnal planetary boundary layer mean height dropping to approximately 201 meters around 06:00 UTC on July 8, 2026.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 31 of 35

Nighttime stagnation and suppressed mixing strongly support local NO_2 accumulation at the anomaly time because observed wind speed at the nearest ASOS station was 0.0 m/s at 05:55 UTC, the 6-hour mean ASOS wind speed was only 0.3568 m/s for 07-08 06Z, humidity was very high near 88.7 percent to 90.93 percent, and GFS planetary boundary layer height was only about 327.9 m at 06:00 UTC with a low 6-hour mean near 201.3 m.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 32 of 35

The anomaly occurred on July 8th, around 06:00 UTC, and was located approximately 19.067 km from the query point.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 33 of 35

The air quality anomaly in Houston, Texas is caused by high levels of NO_2 particles.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 34 of 35

TCEQ monitoring in Houston recorded a severe nitrogen dioxide anomaly of 7.5 ppb at 06:00 UTC on July 8, 2026, significantly higher than the expected baseline value of 1.12 ppb.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 35 of 35

A distinct industrial or ship-channel plume has weaker direct support in the available data because TCEQ SO2 near the period remained near zero or negative, Sentinel-5P daytime columns did not show an obvious contemporaneous regional enhancement in NO2 or CO before the event, and particulate levels near the anomaly were not unusually high at 06:00 UTC.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.